

Production Data Challenge

Insight & Power BI Research Report

From 20 cleaned data files to a decision-ready model for operational efficiency, data reliability and cost reduction

Challenge statement Combine production, product, stoppage, staffing and energy data to understand packaging performance, reduce energy consumption, improve production efficiency, explain stoppages, and improve future data quality.

20	~993K	2	Power BI
cleaned files	source rows	workstreams	replication-ready

Scope date: 24 September 2026 | Data focus: packaging operations (KM1, KM2, AXO10, AXO11, AXO12)

Source / basis: BBMED Production Data Challenge definition and provided relationship diagram; calculations use the 20 cleaned Excel workbooks supplied with the project.

1. Executive summary

What the board/operations team should take from the current data

Finding	What it means for BBMED
Workstream B is decision-ready	430 MES 2026 orders, 398 Excel orders and 263 matched orders support a strong production/data-consistency analysis.
Most source differences are small, but the tail is operationally important	59.7% of matched orders are within $\pm 1\%$ quantity and 87.1% within $\pm 5\%$; median absolute gap is 0.65%, but p90 is 9.46%.
Stoppage intensity is the strongest operational signal found	Disturbance minutes per usage hour has $r = -0.56$ with gross production rate. This is an association, not proof of causality.
Small orders show more fixed loss	Smallest order quartile: median 10.85 u/min and 27.9 disturbance min per usage hour. Largest quartile: 16.74 u/min and 20.0 min/h.
Staffing cannot yet be optimized from headcount alone	Average workers vs rate has $r = -0.014$. Labor productivity falls as staffing rises, but product complexity and incomplete personnel master data strongly confound the result.
Workstream A is only a pilot with the current energy sample	Energy covers 3.49 hours, 5 KM1 meters and one overlapping order. The meter unit is unspecified and density exists for only 4/209 active 2026 products.

Priority recommendation Build the Power BI model around order/work-record grain, launch a MES ↔ Excel exception dashboard and stoppage Pareto first, then extend energy collection for at least several weeks before making product-level energy or cost claims.

2. Analytical approach

How the 20 sources were turned into a decision framework

Step	Method
1. Understand & inventory	Profile every file, identify grain, candidate keys, timestamps, quantities, machine/product/person fields and source purpose.
2. Validate relationships	Test OrderNumber, WorkRecordID, ItemCode, personnel, stoppage and meter keys; quantify match coverage rather than assuming the relationship PDF is always correct.
3. Standardize keys & definitions	Convert order/article identifiers to text, union 2025/2026 periods, map manual machine codes, build composite stoppage-reason keys, normalize reason labels.
4. Reconcile sources at comparable grain	Aggregate MES and manual Excel to OrderNumber before comparing quantity, start/end timestamps, duration and rate.
5. Build operational KPIs	Throughput, availability proxy, performance, rejects/quality proxy, OEE proxy, disturbance intensity, employee-hours and source variance.
6. Add energy as an interval fact	Convert cumulative readings to interval deltas; allocate each interval to KM1 order/state by timestamp overlap. Do not directly relate raw meter readings to orders.
7. Interpret statistically and operationally	Use correlations and within-product/machine comparisons as association evidence; show sample size, assumptions and alternative explanations.
8. Translate findings into controls	Define Power BI monitoring pages, exception rules, cost formulas, master-data requirements and data-collection improvements.

Research principle The analysis deliberately avoids “one system is correct” conclusions. It measures agreement, isolates exceptions and identifies where definitions or coverage prevent a reliable answer.

3. Data landscape: 20 cleaned tables

Recommended role in the Power BI solution

#	Source table	Rows	Recommended model role
1	dw_energy_meter_x_machine	5	Energy meter dimension / component map
2	dw_energy_readers_data	1,000	Raw cumulative energy readings
3	erp_product_info	5,589	Product dimension; density / pack attributes
4	excl_production_data_2026	1,666	Manual production fact / comparison source
5	mes_cycle_data	627,876	Raw cycle/event log; hidden staging / drill-through
6	mes_stoppages	43,848	Stoppage fact by WorkRecordID
7	mes_stoppage_types	10	Stoppage type dimension
8	mes_stoppage_reason_per_machine_group	200	Reason dimension; composite key required
9	mes_stoppage_categories	2	Reference taxonomy; no direct usable key in current fact
10	mes_shift_info	999	Shift dimension after removing 997 blank rows
11	mes_temp_stoppages	272,552	Raw status/event table; energy-state and audit layer
12	mes_rejections_ids_x_work_ids	312	Reject fact
13	mes_machine_groups	2	Machine-group dimension
14	mes_production_periods_2025	3,240	Production-period fact (union)
15	mes_production_periods_2026	2,783	Production-period fact (union)
16	mes_person_x_work_record	30,958	WorkRecord ↔ Person bridge
17	mes_item_prod_rate	1,002	Product rate enrichment; deduplicate
18	mes_production_orders	621	Order dimension / planned attributes
19	mes_prod_order_data	431	Order audit snapshot / validation source
20	mes_personnel	44	Personnel dimension

Source / basis: Row counts are data rows excluding header. Total across sources = 993,140 rows.

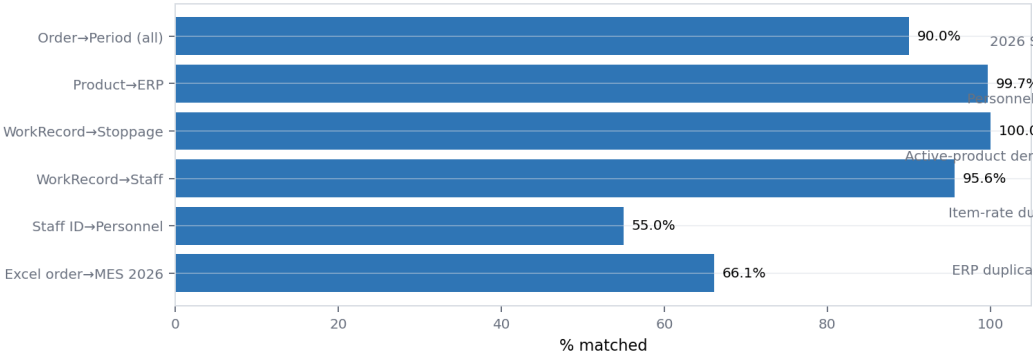
4. Data quality, completeness & relationship assessment

Data quality & relationship readiness

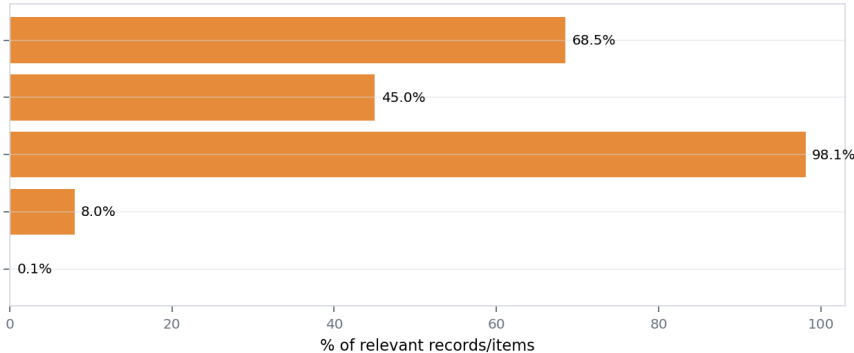
20 cleaned files profiled; relationships tested at business-key level



Key relationship coverage



Highest-priority completeness issues



- Critical** Energy unit is unspecified and only 3.49 hours are present; density exists for only 4 of 209 active 2026 products.
- High** Shift dimension contains only 2 non-empty rows while 68.5% of 2026 production periods have no Shift value.
- High** Only 33 of 60 personnel IDs appearing in the work-record bridge are found in the personnel master.
- Medium** ERP has 6 duplicate ItemCode rows; standard-rate table has 80 duplicate rows and needs a deterministic dedup/conflict rule.
- Definition** Stoppage reasons include RUHE plus a separate Ruhe state, and Pause/PAUSE variants; planned vs unplanned loss must be normalized.

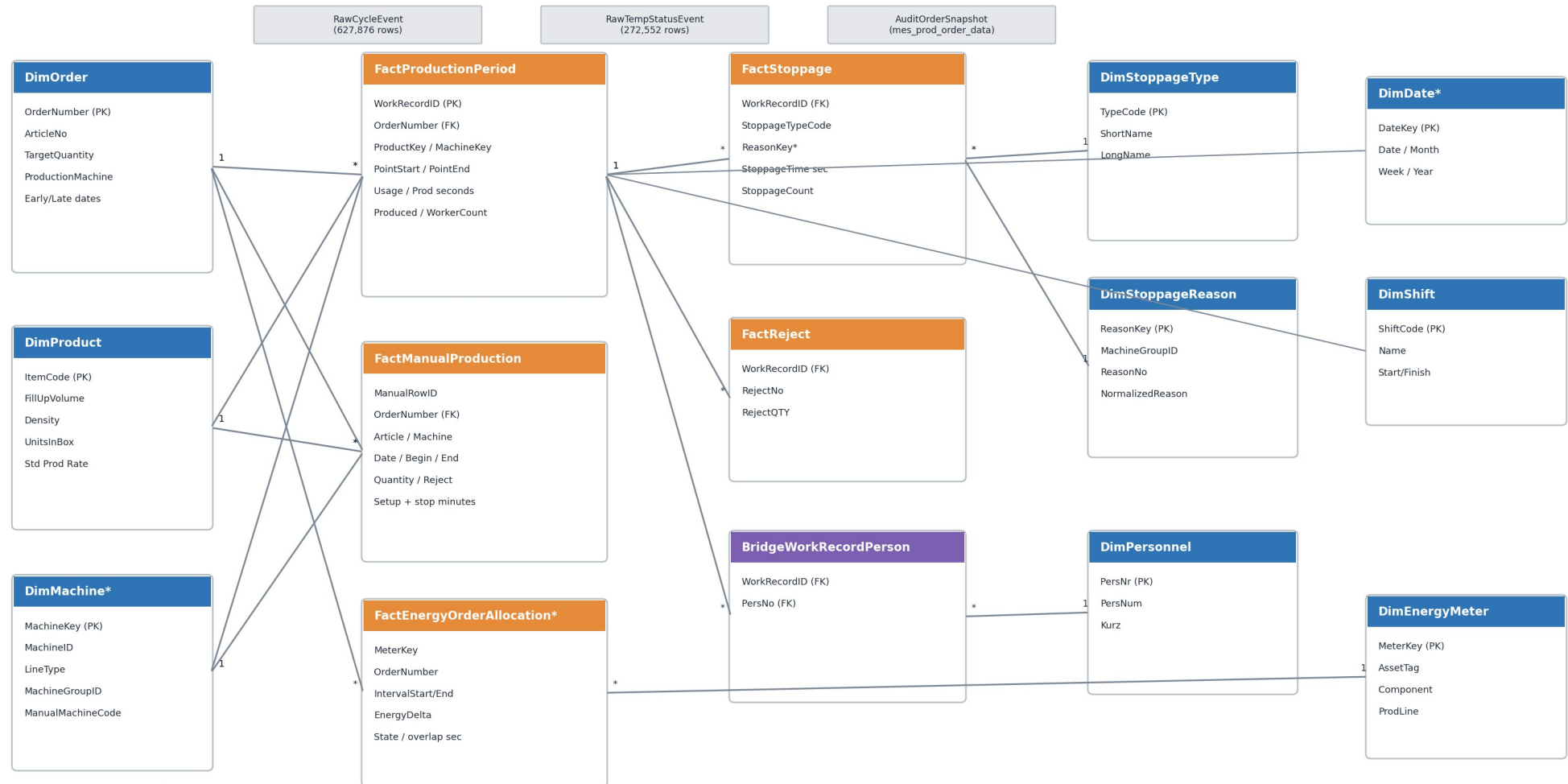
Key interpretation: the product/order/work-record backbone is strong, while shift, personnel master, density and energy metadata are the main blockers. These issues directly affect OEE, staffing, energy-per-unit and cost calculations.

5. Recommended Power BI data model

Power BI-style semantic model based on the tested relationships

Recommended Power BI semantic model

Star/constellation design - active relationships single-direction; raw event logs kept as staging/audit



* Derived table / column in Power Query. Energy-to-order is a range/timestamp allocation table, not a direct relationship.

Modeling rule: keep relationships single-direction from dimensions/header facts into child facts. Do not connect raw cycle/status tables everywhere; use them as staging/drill-through to avoid ambiguous filters and model bloat.

6. KPI framework

What should be shown and how it should be calculated

KPI	Calculation	Why it matters
Produced Units	$\Sigma \text{ ProducedDuringThePeriod}$	Core output volume; compare by order/product/machine.
Usage Hours	$\Sigma \text{ CalcUsageTime} / 3600$	Gross time denominator.
Production Hours	$\Sigma \text{ CalcProductionTime} / 3600$	Apparent productive operation.
Gross Production Rate	$\text{Produced Units} / \text{Usage Minutes}$	Best operating throughput KPI because it includes operational losses.
Net Run Rate	$\text{Produced Units} / \text{Production Minutes}$	Machine running speed while productive.
Availability Proxy	$\text{Production Hours} / \text{Usage Hours}$	Proxy only; planned production time is not separately defined.
Performance	$\text{Produced} / \Sigma (\text{TargetRate} \times \text{ProductionMin})$	Target-speed attainment; flag >120%.
Quality Proxy	$\text{Produced} / (\text{Produced} + \text{RejectQTY})$	Assumes MES produced count excludes rejects; validate.
OEE Proxy	$\text{Availability} \times \text{Performance} \times \text{Quality}$	Trend KPI, not certified OEE until definitions are confirmed.
Disturbance Minutes	$\text{Type 1 stoppage_time} / 60$	Use reason and machine group for Pareto.
Disturbance Intensity	$\text{Disturbance Min} / \text{Usage Hours}$	Comparable across different order durations.
Employee Hours	$\Sigma \text{ WorkerCount} \times \text{UsageSec} / 3600$	Labor input.
Units / Employee Hour	$\text{Produced} / \text{Employee Hours}$	Labor productivity, only after staff completeness checks.
MES ↔ Excel Qty Delta %	$(\text{MES Qty} - \text{Excel Qty}) / \text{Excel Qty}$	Primary data-reliability exception KPI.
Start-Time Gap	$\text{ABS}(\text{MES first start} - \text{Excel first start})$	Detect recording-period misalignment.
Energy Delta	$\text{Current cumulative} - \text{Previous cumulative}$	Per meter interval; flag reset/negative values.
Energy / 1,000 Units	$\text{Energy in same time window} / \text{units in same time window} \times 1000$	Never use full-order quantity against a partial energy window.
Cost of Loss	$\text{Loss quantity/time/energy} \times \text{validated cost rate}$	Use What-If parameters for labor, line-hour and electricity cost.

7. Power BI page 1 - Executive Operations Dashboard

Packaging Operations - Executive View

2026 MES production periods; calculated from cleaned challenge data

Produced units

11.63 M

430 MES orders

Median gross rate

14.2 u/min

usage-time denominator

Median availability proxy

42.4%

productive / usage time

MES↔Excel within ±5%

87.1%

229 / 263 matched

Disturbance-rate correlation

-0.56

vs gross production rate



- 1

Small orders carry more fixed loss
Q1 orders show 10.85 u/min median and 27.9 disturbance min per usage hour; Q4 orders reach 16.74 u/min and 20.0 min/h.
- 2

Most MES/Excel records are close - but outliers are severe
Median absolute quantity gap is 0.65%, yet the 90th percentile reaches 9.46%; exception management matters more than blanket replacement of one source.
- 3

Stoppage intensity is strongly associated with lower throughput
Across valid 2026 orders, disturbance minutes per usage hour and gross rate have $r = -0.56$ (association, not causation).

Recommended slicers: Year/Month, Machine, Product, Order size band, Shift and Source Quality Flag. Add drill-through from any order to work records, stoppages, staffing and source reconciliation.

8. Workstream B - MES vs Excel consistency

MES vs Excel - Data Consistency

Order-level comparison after aggregating both sources by production order

Matched orders

263

of 398 Excel orders

Within $\pm 1\%$ quantity

59.7%

157 orders

Within $\pm 5\%$ quantity

87.1%

229 orders

Median abs quantity gap

0.65%

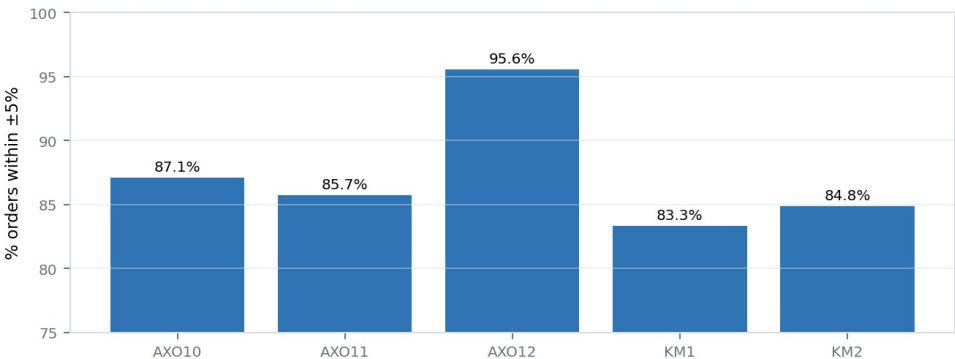
134 units median abs gap

Median start-time gap

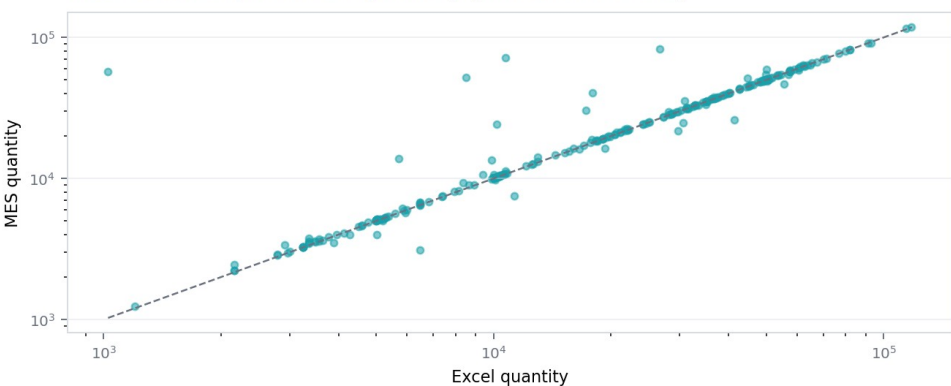
12.8 min

p90 = 4.18 h

Quantity consistency by MES machine



MES vs Excel produced quantity (matched orders)



Largest quantity-difference outliers

Order	Machine	MES Qty	Excel Qty	Delta %	MES dur (h)	Excel dur (h)
701145	AXO11	57,554	1,026	+5509.6%	42.4	2.1
701045	KM2	71,850	10,746	+568.6%	145.6	12.9
701753	KM2	51,793	8,500	+509.3%	158.8	8.6
701666	KM1	82,448	26,640	+209.5%	71.6	25.5
701439	AXO10	13,896	5,700	+143.8%	16.4	6.6

Interpretation: quantity and timestamp agreement is generally close, but a small number of orders have order-closure or time-window mismatches large enough to distort production rate. The dashboard should therefore focus on exception detection, not on choosing one source globally.

9. Production performance & OEE proxy

Production Performance & OEE Proxy

OEE is provisional because planned-production-time and reject semantics require validation

Usage time

16,280 h

sum CalcUsageTime

Productive time

5,921 h

sum CalcProductionTime

Median availability

42.4%

production / usage

Median performance

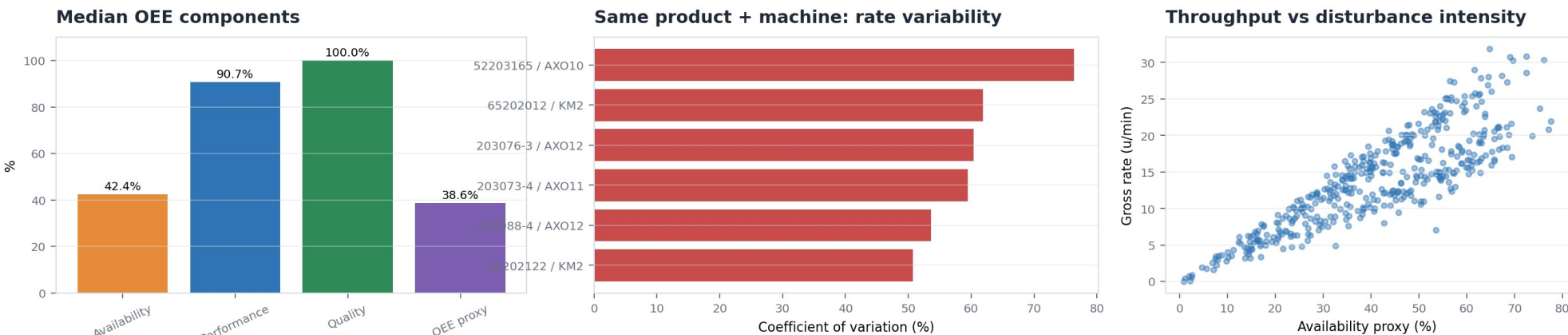
90.7%

vs time-weighted target

Median OEE proxy

38.6%

$A \times P \times Q$



- 21 of 429 orders have performance >120%, indicating target-rate or period-definition anomalies.
- 381 of 429 OEE-eligible orders have no linked rejection quantity; treat Quality=100% as provisional, not confirmed zero defects.
- Use OEE for trending only after BBMED validates planned production time, good-count/reject semantics, and target-rate master data.

Operational finding: performance varies strongly even for repeated Product × Machine combinations. This makes same-product/same-machine benchmark bands more useful than comparing machine averages across different product mixes.

10. Stoppage causes and lost-time logic

Stoppage / Loss Analysis

Type 1 records treated as disturbance events; reason normalization is required before management reporting

Disturbance time

8,145 h

Type code 1, 2026

Top reason: RUHE

3,355 h

41.2% of disturbance

Rüsten / setup

2,052 h

25.2%

Cleaning

769 h

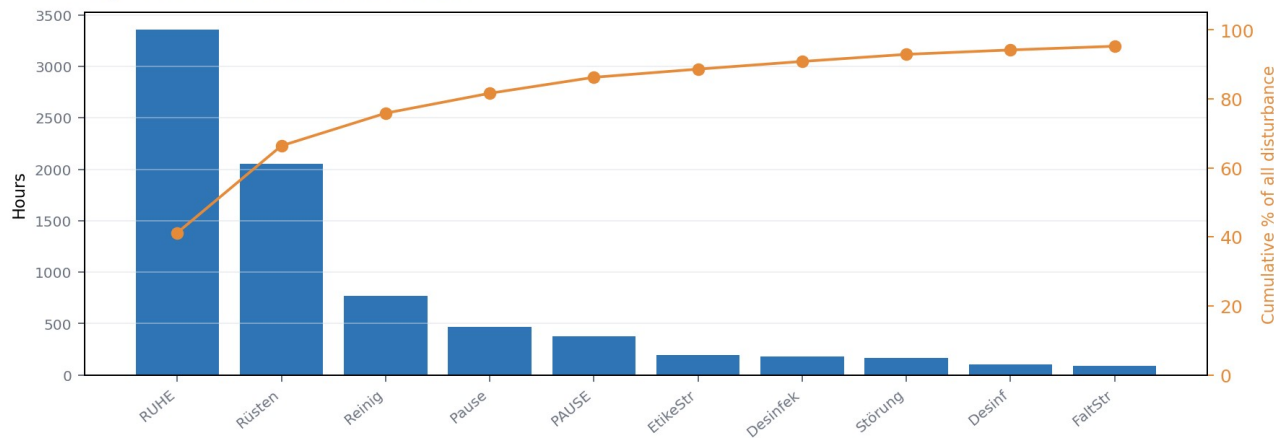
9.4%

Rate association

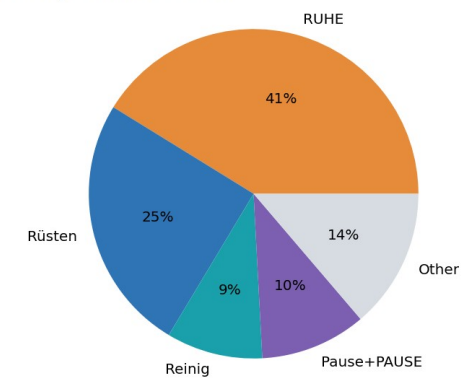
$r = -0.56$

disturbance intensity vs rate

Top disturbance reasons - 2026



Definition problem to fix



Interpretation

RUHE is recorded as a disturbance reason while 'Ruhe' also exists as a stoppage/state type. Pause and PAUSE are separate labels across machine groups. Planned setup, cleaning and breaks should be split from unplanned technical losses before assigning cost or accountability.

Cost logic to add: Lost Capacity Units = Disturbance Minutes × comparable benchmark gross rate. Convert to € only after BBMED supplies line-hour/contribution rates and the planned/unplanned reason taxonomy is validated.

11. Staffing level vs production conditions

Staffing & Labor Productivity

Aggregate staffing relationships are confounded by product, order size, and data completeness

WorkRecord staffing coverage

95.6%

5,759 / 6,023 WRs

Personnel master coverage

55.0%

33 / 60 staff IDs

Median units / employee-hour

296

weighted WorkerCount

Workers vs rate correlation

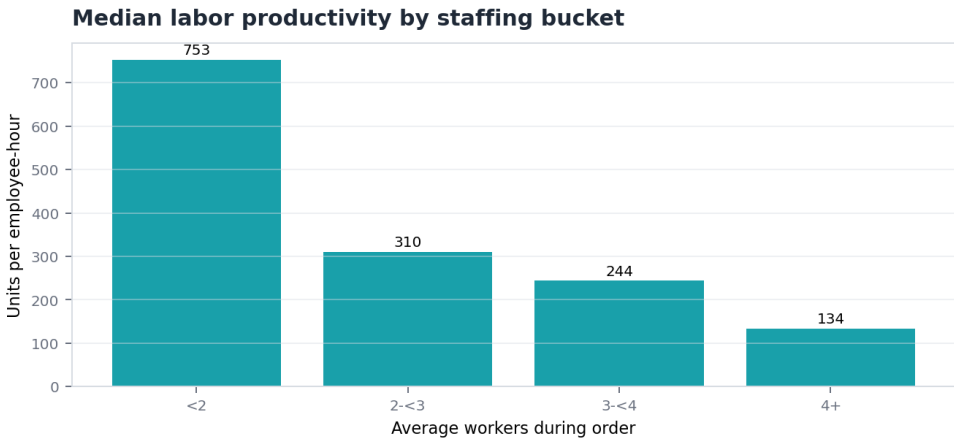
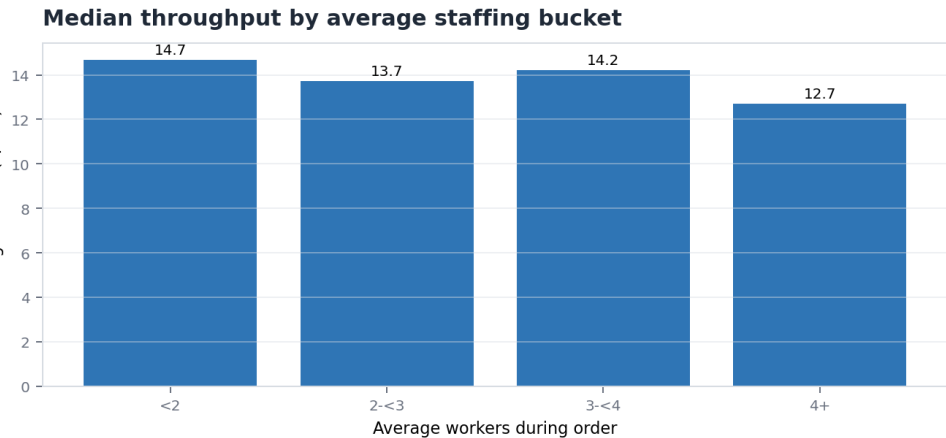
-0.014

no aggregate linear effect

Workers vs labor productivity

-0.61

strong confounding / diminishing-return flag



How to use this in Power BI

Compare staffing only within the same Product × Machine × Order-size band. Show rate, disturbance intensity and employee-hours together; flag orders where extra staffing does not improve throughput.

Do not use the current aggregate buckets to reduce headcount directly: high-complexity jobs may require more people, and 45% of staff IDs are not resolved in the personnel master.

Recommended management use: benchmark employee-hours only inside comparable Product × Machine × Order-size groups, then investigate orders with above-benchmark labor input and no throughput benefit. Do not infer headcount reductions from the aggregate relationship.

12. Workstream A - KM1 energy pilot

KM1 Energy - What the Current Data Can and Cannot Support

Observed window: 23 Sep 2026 14:26:40 to 17:56:18; cumulative meter unit is not defined

Observed duration

3.49 h

single short window

Energy meters

5

all mapped to KM1 components

Total cumulative delta

11.227 units

unit must be confirmed

Active order

702026

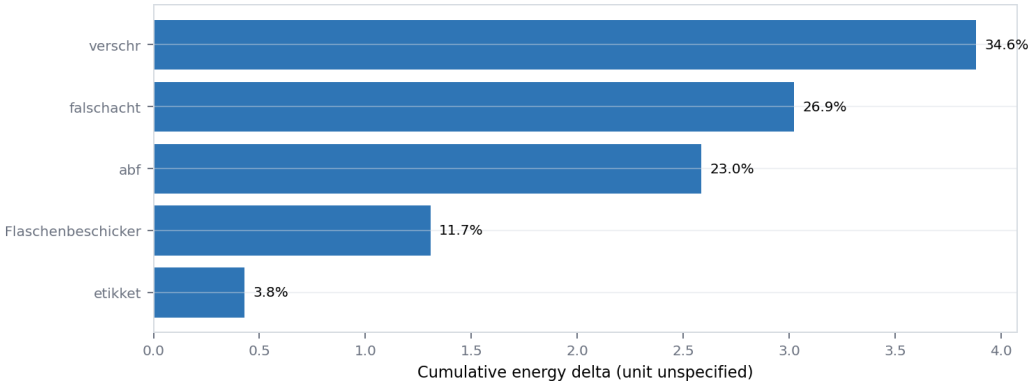
product 10269202061-1

Non-prod / unknown state share

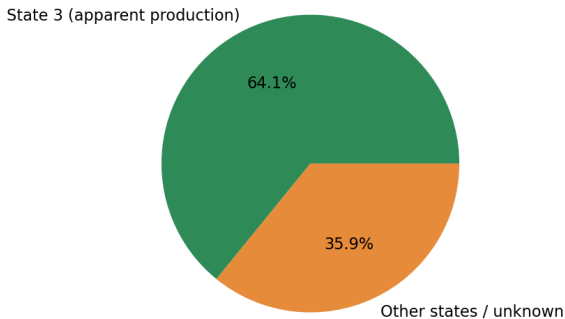
35.9%

provisional state inference

Energy delta by KM1 component



Observed-window energy by apparent machine state



Observed-window production denominator

Temp status data shows 0 units before the energy window and 3,165 units at the last event inside it; the first event after the window shows 3,608 units. This implies a provisional intensity range of about 3.1-3.5 energy-units per 1,000 produced units.

Decision limit

Only one order overlaps the energy data, product density is missing for this item, and the energy unit is unspecified. Therefore product grouping, density correlations, order ranking, and cost savings cannot yet be stated reliably.

The current energy data can validate the technical allocation method, but it cannot support reliable product ranking, density correlation or annual savings. Extend the same interval model to a longer data window before operational decisions.

13. Power BI calculations and Power Query transformations

Replication-ready logic for the dashboard build

Core measures	Loss / comparison measures
Produced Units MES Produced Units = SUM(FactProductionPeriod[ProducedDuringThePeriod])	Disturbance Min Disturbance Min = CALCULATE(DIVIDE(SUM(FactStoppage[StoppageTime]),60), KEEPFILTERS(DimStoppageType[TypeCode]=1))
Usage Hours Usage Hours = DIVIDE(SUM(FactProductionPeriod[CalcUsageTime]), 3600)	Disturbance Intensity Disturbance Min / Usage Hr = DIVIDE([Disturbance Min],[Usage Hours])
Gross Rate Gross Rate u/min = DIVIDE([MES Produced Units], DIVIDE(SUM(FactProductionPeriod[CalcUsageTime]),60))	Manual Qty Excel Produced Units = SUM(FactManualProduction[Quantity])
Availability Availability Proxy = DIVIDE(SUM(FactProductionPeriod[CalcProductionTime]), SUM(FactProductionPeriod[CalcUsageTime]))	Qty Delta % Qty Delta % = DIVIDE([MES Produced Units]-[Excel Produced Units],[Excel Produced Units])
Theoretical Output Theoretical Output = SUMX(FactProductionPeriod, FactProductionPeriod[TargetUnitsPerMinute] * FactProductionPeriod[CalcProductionTime] / 60)	Units / Employee Hr Units per Employee Hour = DIVIDE([MES Produced Units],[Employee Hours])
Performance Performance = DIVIDE([MES Produced Units], [Theoretical Output])	Energy Energy Units = SUM(FactEnergyOrderAllocation[EnergyDelta])
Quality Quality Proxy = DIVIDE([MES Produced Units], [MES Produced Units] + [Reject Qty])	Energy / 1k Energy per 1000 = DIVIDE([Energy Units],[Units During Energy Window]) * 1000
OEE OEE Proxy = [Availability Proxy] * [Performance] * [Quality Proxy]	Energy Cost Energy Cost € = [Energy kWh] * SELECTEDVALUE(pElectricityPrice[EUR_per_kWh])
Employee Hours Employee Hours = SUMX(FactProductionPeriod, FactProductionPeriod[WorkerCount] * FactProductionPeriod[CalcUsageTime] / 3600)	Loss Cost Loss Cost € = [Lost Capacity Hours] * SELECTEDVALUE(pLineCost[EUR_per_Hour])

Power Query must do the heavy data engineering Union 2025/2026 periods; enforce text keys; deduplicate product/rate masters with conflict flags; map manual machine codes; create ReasonKey = MachineGroupID & "|" & ReasonNo; build energy interval deltas and timestamp-overlap allocation before DAX.

14. How the analysis translates into operational & processing cost reduction

Lever	Operational action	KPI / formula	Cost link
Reduce fixed loss on small orders	Campaign similar products, sequence orders to reduce cleaning/setup, compare Q1 orders against same-product-machine benchmark.	Opportunity Hours = Actual Usage Hours - Produced / Benchmark Rate / 60	Line-hour cost × opportunity hours; plus avoided energy during nonproductive time.
Attack high-impact stoppage reasons	Separate planned setup/cleaning/breaks from unplanned technical stops; Pareto by hours and frequency.	Lost Capacity Units = Disturbance Min × Benchmark Rate	Lost contribution / line-hour cost; maintenance prioritization.
Control MES ↔ Excel exceptions	Automatically flag quantity >±1%, >±5%, start/end >60 min, duration mismatch and unmatched orders.	Exception Count / Exception % / Rework Hours	Less manual reconciliation, fewer incorrect KPIs and faster month-end reporting.
Optimize staffing by comparable work	Use Employee Hours and Units/Employee Hour within Product×Machine×Order-size bands.	Excess Employee Hours = Actual - Benchmark for comparable orders	Loaded labor rate × excess hours; only after personnel master completeness improves.
Reduce nonproductive energy	Measure energy by machine state and stoppage reason; focus on long idle/setup states that can safely power down.	Nonproductive kWh × tariff; kWh/1k units	Electricity savings and lower energy intensity.
Improve master data at source	Require shift, product density, status taxonomy, meter unit and valid personnel IDs.	DQ score / unresolved-key count / missing-field rate	Prevents recurring cleansing and protects KPI trust.

Best first operational hypothesis to test Small orders appear to suffer more disturbance intensity and lower rate. Test campaign scheduling / faster changeover on selected products, then compare against a same-product-machine historical benchmark rather than a plant-wide average.

15. Validation rules & future data governance

Rules that should be implemented in source systems / Power BI quality gates

Control area	Validation rule
Keys	OrderNumber and WorkRecordID required; Reading ID unique; ItemCode stored as text; no silent truncation of hyphenated codes.
Referential integrity	Every production period must resolve to order/product/machine; every staff ID should resolve to personnel; unresolved keys shown as DQ exceptions.
Time	PointEnd $\geq$ PointStart; duration matches timestamp difference within tolerance; manual Begin/End valid; overnight shifts handled explicitly.
Quantity	Produced/reject quantities nonnegative; cumulative event quantity cannot decrease without a documented reset; target and actual grain separated.
Rate	Target rate must come from one approved master; performance >120% is a warning; duplicated item-rate rows require an effective-date or priority rule.
Stoppage	Type and reason required; planned/unplanned category mandatory; normalize Pause/PAUSE and overlapping Ruhe definitions while retaining source label.
Staffing	WorkerCount ≥ 0 ; bridge staff IDs valid; shift required; staff intervals should have join/leave timestamps for true employee-hours.
Energy	Meter unit explicitly kWh (or other); cumulative reset flag; sampling gaps monitored; meter $\rightarrow$ component mapping effective-dated; order/state overlap stored.
Product master	Density, fill volume and packaging attributes required for KM1 items used in energy analysis.
Reconciliation	MES vs Excel quantity, time and duration exceptions stored with reason/owner/resolution rather than overwritten.

Data-quality KPI page Show % valid keys, % complete shift/personnel/product attributes, unmatched orders, duplicates, energy sampling gaps, status-definition violations and reconciliation exceptions as first-class KPIs alongside production performance.

16. Power BI implementation roadmap

Build the reliable operational model first; then scale energy

Phase	Timing	Build activities	Deliverable
Phase 1 - Foundation	1-2 weeks	Power Query staging; union periods; canonical keys; model relationships; Date/Machine dimensions; DQ rules.	Validated semantic model + DQ page
Phase 2 - Workstream B	1-2 weeks	MES ↔ Excel order reconciliation; production-rate/OEE proxy; stoppage Pareto; staffing bridge; drill-through.	Executive + consistency + operations pages
Phase 3 - Cost layer	1 week	What-If parameters for line €/h, labor €/h, electricity €/kWh; benchmark-based opportunity calculations.	Cost opportunity page
Phase 4 - Energy expansion	4-8 weeks data collection	Confirm unit; collect continuous KM1 readings across many orders; capture status dictionary and density; validate interval allocation.	Reliable energy/order/product KPIs
Phase 5 - Operations adoption	Ongoing	Exception ownership, weekly review, reason taxonomy cleanup, threshold tuning, incremental refresh and audit trail.	Sustained KPI trust and savings tracking

Performance recommendation Do not import all 900K+ event rows into every report visual. Use incremental refresh / aggregation tables for RawCycleEvent and RawTempStatusEvent, and expose detail only on drill-through pages.

17. Assumptions, limitations & questions that cannot yet be answered reliably

Limitation	Impact on interpretation
Energy unit	"Total Energy Passed" has no unit metadata. Energy cost and kWh KPIs must remain disabled/provisional until confirmed.
Energy sample	Only 3.49 hours and one overlapping KM1 order. No reliable product ranking, density correlation, order-size comparison or annual savings extrapolation.
Density	Only 4 of 209 product codes used in 2026 production periods have a usable density field.
OEE definition	Planned production time is not explicitly separated. Availability is therefore an operational proxy based on <code>CalcProductionTime / CalcUsageTime</code> .
Quality	Most orders have no rejection record; absence may mean zero rejects or missing capture. Quality=100% is not automatically trustworthy.
Stoppage taxonomy	Type 1 reason labels mix planned setup/cleaning/breaks and apparent idle states with technical faults; reason-level "waste" needs classification.
Machine groups	AXO → group 2 is clear; mapping KM1/KM2 → group 1 is inferred from available tables and should be confirmed, particularly KM2.
Staff master	27 of 60 staff IDs in the work-record bridge do not resolve to the provided personnel master.
Shift	2026 production-period Shift is missing in 68.5% of rows, limiting shift comparison.
Causality	Correlations do not prove that stoppages, staffing or order size cause the performance difference; product mix and scheduling are alternative explanations.

Additional data required Continuous KM1 energy history, meter unit/tariff, complete product density/pack attributes, documented status dictionary, planned-production calendar, reliable reject semantics, complete personnel master and explicit planned/unplanned stoppage category.

18. Detailed findings and evidence strength

Observations, calculation logic, support level and action

Finding	Observed result	Support	Evidence	Operational use
MES ↔ Excel quantity consistency	Median absolute gap 0.65%; 87.1% within ±5%; p90 9.46%.	263 matched orders	High	Build threshold-based exception workflow; investigate tail orders.
Timestamp consistency	Median absolute start gap 12.8 min; 68.1% within 30 min; p90 4.18 h.	263 orders	High	Separate minor operator timing offsets from order-boundary errors.
Stoppage intensity vs rate	$r = -0.56$ between disturbance min/usage h and gross rate.	~429 valid orders	Medium	Prioritize high-duration/high-frequency loss reasons; validate causality with controlled comparisons.
Order size vs rate	$r = +0.24$; Q1 median 10.85 u/min vs Q4 16.74 u/min.	429 orders	Medium	Test campaign scheduling / changeover reduction for small batches.
Staffing vs rate	$r = -0.014$ overall; no monotonic throughput gain across staffing buckets.	429 orders	Low-Medium	Use within-product-machine benchmarks, not plant-wide headcount targets.
Repeated-product variability	Several Product×Machine groups show CV >50% in gross rate (e.g., 10269203088-4 on AXO12: 8 orders, CV ≈53.6%).	Groups with ≥4 orders	Medium	Create expected-rate bands and root-cause outlier orders.
Energy component share	During the 3.49 h sample: Verschrauber 34.6%, box-packing 26.9%, filler 23.0%, feeder 11.7%, labeller 3.8%.	5 meters × 200 readings	Pilot only	Use as instrumentation validation, not annual optimization ranking.
Observed energy state split	Apparent state 3 accounts for 64.1% of energy; other/unknown states 35.9%.	One order/window	Pilot only	Confirm status dictionary, then separate productive/idle/setup energy.

19. Appendix - relationship mapping & transformation assumptions

Parent / key	Child / key	Cardinality	Transformation / note
mes_production_orders.OrderNumber	mes_production_periods.OrderNo	1:*	Core order → work-record relationship; tested.
mes_production_orders.OrderNumber	mes_cycle_data.Auftrag	1:*	Raw event drill-through; keep staging/hidden.
mes_production_orders.OrderNumber	mes_temp_stoppages.production_order_number	1:*	Status/event audit and energy-state allocation.
mes_production_orders.ArticleNo	erp_product_info.ItemCode	*:1	Standardize both as text; ERP has 6 duplicate ItemCode rows to resolve.
mes_production_orders.ArticleNo	mes_item_prod_rate.Art_Nr	*:1 expected	80 duplicate rate rows require rule/effective date.
mes_production_periods.WorkRecordID	mes_stoppages.work_record_id	1:*	Complete work-record coverage in supplied period data.
mes_production_periods.WorkRecordID	mes_person_x_work_record.WorkRecordId	1:*	Staff bridge; 95.6% WR coverage.
mes_production_periods.WorkRecordID	mes_rejections...WorkRecordId	1:*	Reject detail; sparse by order.
mes_personel.PersNr	mes_person_x_work_record.PersNo	1:*	Only 33/60 bridge IDs resolve to personnel master.
mes_shift_info.shift_code	period/cycle/temp shift	1:~ expected	Shift master has only 2 active rows; 2026 period shift missing heavily.
mes_stoppage_types.id	mes_stoppages.stoppage_type_code	1:*	Codes 1-3 observed; type 1 used for disturbance analysis.
machine_groups + reason number	stoppage reason	1:*	Create composite ReasonKey; AXO → group2, KM → group1 inference.
energy meter name_new	energy reading asset tag	1:*	5/5 meters mapped to KM1 components.
Excel Auftragskarte Nr.	MES OrderNumber	*:1	Aggregate Excel to order first; 263 matched 2026 orders.
Excel Artikelnummer	ERP ItemCode	*:1	Standardize text to preserve suffixes like "-1".
Energy interval timestamps	Production-period/status timestamps	range allocation	Create FactEnergyOrderAllocation; not a direct physical relationship.

Source / basis: Relationship basis: provided "Relatnship between table.pdf"; cardinalities and model roles were refined by profiling and join-coverage checks against the cleaned workbooks. Challenge requirements: provided BBMED Production Data Challenge definition.